

Feature request - Mode 3: Win95-era thin moot CLI (2026-09-19)

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Status: parked for build agent

Repo: SimonBarnett/agentic_irc

Related: Mode 1 (agent->agent SEAL), Mode 2 (moot + FILE), Mode DUMB (Server 2012 `dumb\_agent.py` / deferred .NET Phase 5). Mode 3 is **not** a reimplement of Phase 5; it is a new product surface.

One-line goal

Ship a **single native Windows console EXE** that a human runs on ancient-through-modern Windows. It **joins a moot** as a thin command-line client so moot agents can **send shell commands** and **receive stdout/stderr**, then publish that EXE as a **GitHub Release** that **rebuilds update**.

Why (operator story)

1. Operator copies `airc-moot-thin.exe` (and a tiny config / key drop) onto a box that may be as old as **Windows 95** (exclude Windows ME).
2. Operator runs it; it connects to Libera TLS, joins the private channel, and **JOINS** the named moot as a non-LLM participant.
3. A chair/agent with floor (or an operator allowlist - see LOCKED vs UNKNOWN) sends a command.
4. The thin client runs the command under a **jail + allowlisted bins**, returns **stdout/stderr/rc** into the moot (or DUMB-shaped sealed result - see protocol).
5. Rebuilds of the same artefact land on the **same GitHub Release** (tag `mode3-thin` or `airc-moot-thin-v`) so operators always download the latest known-good binary.

LOCKED

| ID | Rule |

|---|---|

| L1 | **Backwards compatibility bar:** one Win32 **ANSI** PE (console subsystem) that is intended to run on **Windows 95** OSR2, 98, NT 4, 2000, XP, Vista, 7, 8, 10, 11, Server 2003-2022. Explicitly **exclude Windows ME**. Document any API that forces a higher floor. |

| L2 | **Not an LLM.** No model, no Grok, no Python runtime required on the target box. |

| L3 | **Moot membership.** Client must OPEN/JOIN (or JOIN only if chair already opened) a moot id supplied by config/CLI; appear on roster; participate as thin CLI (SAY of results and/or sealed job replies). |

| L4 | **Command relay.** Agents (or allowlisted operators) can pass a command; thin client executes and returns **stdout, stderr, exit code**. Truncation must be honest (truncated: true + spill file under jail if over cap). |

| L5 | **Jail + allowlist.** Same spirit as DUMB: `--allow-path` jail, default bins (cmd.exe, hostname.exe, ipconfig.exe, whoami.exe; PowerShell only when OS supports it). Unknown operator / empty allowlist -> **no exec**, no result ciphertext on the wire. |

| L6 | **Secrets off-channel.** PSK / connector key generated off IRC; never printed; fingerprint compare OOB. No `password=` or API key assignments in repo or release notes. |

| L7 | **GitHub Release.** Binary (and checksums) published under `SimonBarnett/agentic\_irc` Releases. **CI** rebuild updates that release (replace assets on a stable tag, or move `mode3-thin-latest` + keep versioned tags - pick one in Phase 0 and stick to it). |

| L8 | **Offline tests first.** CI must not open Libera. Protocol/unit tests on a modern builder; ancient OS smoke is manual or VM matrix documented in `docs`. |

| L9 | **Do not break** existing Mode 1/2 Python `irc\_agent` / moot / FILE / DUMB Python paths unless marked compatible extension. |

| L10 | Artefact name: **airc-moot-thin.exe** (plus .sha256). Optional companion .ini / .cfg example in release zip. |

UNKNOWN (Phase 0 must close before claiming Win95 works)

| ID | Question | Exit criteria |

|---|---|---|

| U1 | Can Libera **TLS 1.2+** be spoken from Win95 without a modern Schannel? | Spike: static mbedTLS/LibreSSL vs external relay. Document floor OS if Win95 cannot do TLS. |

| U2 | Is "DOS program" literal **16-bit real-mode DOS**, or **Win32 console that feels like DOS**? | Default assumption until closed: **Win32 ANSI PE** (L1). If Simon insists on true DOS, need a DOS TCP+TLS stack plan or a userspace relay on a newer hop. |

U3	Wire shape: reuse ****DUMB v1**** sealed jobs inside moot, or new ****MOOT-CLI v1**** verbs?	Prefer reuse DUMB job JSON (``exec`/`ping`/`sysinfo``) + moot presence, unless moot floor semantics demand new verbs.
U4	Who may exec: moot floor holder only, chair only, or ``--operators`` nick list (DUMB-style)?	Security default: ****operators allowlist****; floor alone is not enough. Confirm with Simon if softer.
U5	Toolchain on CI: OpenWatcom / MinGW-w64 with ``-municode`` off / MSVC with ``WINVER=0x0400``?	One primary toolchain documented; second is optional.
U6	Release strategy: mutable tag ``mode3-thin`` vs immutable ``airc-moot-thin-vX.Y.Z`` + ``latest``?	Document in ``docs/mode3-release.md``.

Non-goals (this FR)

- Completing deferred ****NET 4.5 Phase 5**** ``airc-dumb.exe`` stub (separate ticket).
- GUI, service wrapper, or auto-update beyond "download new release".
- Running on Windows ME.
- Opening Libera from GitHub Actions.
- Shipping Python/Node on the ancient host.

Acceptance (first shippable)

ID	Criterion
A1	``airc-moot-thin.exe`` builds on CI (or documented one-box build) and attaches to a GitHub Release on ``agentic_irc``.
A2	Rebuild of main (or release workflow) ****updates**** that release's assets + checksums.
A3	Offline tests cover: jail refuse, unknown operator drop, exec stdout/rc framing, truncation flag, config parse.
A4	Manual doc: how to run on XP+ today; honest matrix row for Win95/98/NT4 (pass / fail / blocked on U1).
A5	Live smoke on IONOS (modern Windows): join moot with ``cm-bob`` chair + thin client nick; exec ``hostname``; stdout visible to chair.
A6	Bob hostile MRB may pass ****ready for human UAT**** only when A1-A5 hold and U1-U4 are closed or explicitly waived in ``/docs``.

Suggested tree

```
src/moot_thin/      # C (or C++) Win32 ANSI sources
  main.c
  irc_tls.c         # TLS + IRC (or stub + spike notes)
  moot.c
  dumb_job.c        # reuse DUMB job shapes if U3 says so
  jail.c
  README.md         # WINVER, toolchain, known OS floors
.github/workflows/mode3-thin-release.yml
docs/feature-request-mode3-win95-thin-moot-cli-2026-09-19.md # this file
docs/feature-request-mode3-win95-thin-moot-cli-2026-09-19.pdf
docs/build-and-test-plan-mode3.md
docs/mode3-release.md # after U6
tests/test_moot_thin_*.py # protocol fixtures / golden vectors (offline)
```

Kickoff for build agent

1. Park/confirm this FR + write ``docs/build-and-test-plan-mode3.md`` executable without more questions.
2. Close U2/U3/U4 in the plan with ****stated defaults**** (Win32 ANSI; DUMB jobs + moot JOIN; ``--operators``).
3. Spike U1 on paper + smallest linkable TLS; if Win95 TLS impossible, document ****minimum OS**** and still ship XP+/modern binary + release pipeline (do not silently claim Win95).
4. Implement minimal viable thin client against existing Python moot/DUMB for IONOS smoke.
5. Wire GitHub Release workflow (A1-A2).
6. Offline tests (A3). Push. Stop for Bob MRB - do ****not**** claim human UAT.

Relationship to existing dumb connector

	DUMB (Mode "ancient host")	Mode 3 thin moot CLI
Primary host floor	Server 2012 + Python ref / .NET 4.5 stub	Win95->modern Win32 ANSI (excl. ME)
Presence	CAPA on channel	****Moot roster**** + jobs

| Exec model | Sealed DUMB jobs from `--operators` | Same spirit; moot-visible |
| Ship form | `airc-dumb` stub today | **`airc-moot-thin.exe` Release** |

Do not delete or regress `scripts/dumb\_agent.py` while building Mode 3.